

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458290.81615 +/- 0.00085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.039
Transit depth (Rp/Rs)^2	1.77 +/- 1.03 [%]
Orbital Inclination (inc)	83.97 +/- 1.77
Airmass coefficient 1 (a1)	1.204 +/- 0.053
Airmass coefficient 2 (a2)	-0.164 +/- 0.03
Scatter in the residuals of the lightcurve fit is	4.36 %
Best Comparison Star	None
Optimal Aperture	3.71
Optimal Annulus	10.08
Transit Duration (day)	0.08 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.133 ± 0.039 vs 0.13031 (deviation 0.06σ)
Duration fit vs published	0.08 d vs 0.0954 d (deviation 1.17σ)
Mid-time O–C	2.0 min
Depth SNR	2.8
Residual scatter	4.36 %

Light curve

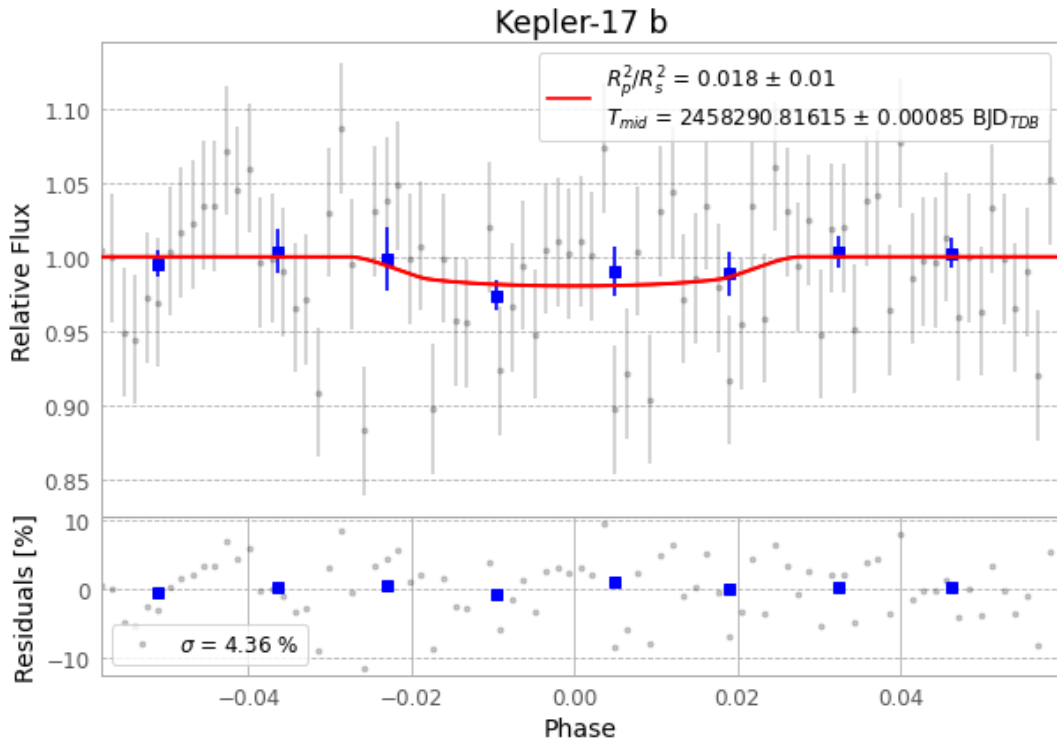


Figure 1 — FinalLightCurve_Kepler-17 b_2018-06-20.png (SHA-256 a589de2a58630e12...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [432, 199], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2bb6cb0b1e796289...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

85 FITS frames (56 MB), KEPLER-17180621052712.FITS → KEPLER-17180621093912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KEPLER-17-180621.log (82fffa9f1778...), FinalLightCurve_Kepler-17 b_2018-06-20.png (a589de2a5863...), FinalLightCurve_Kepler-17 b_2018-06-20.csv (75e6b4e2ef18...)

Compute resources

Wall time 72.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 00:16Z.